Agile Team Charter

SAEON Real Time Rainfall Monitoring System

Reporting period: 28 July to 23 October 2026, Tuesday to Friday.

1. Team purpose and product scope

Our team is developing a rainfall monitoring prototype for SAEON researchers. Remote access to observations is intended to reduce delays and site visits associated with manually retrieving USB data loggers. Environmental researchers are the primary users; conservation organisations, government agencies and communities may benefit from better access to observations.

The ESP32 WROOM-32 reads the TR-525M rain gauge. DHT11 is used for testing; SHT31 is the temperature and humidity sensor selected for the final product. Firebase is the database platform.

The planned cellular data route is ESP32 → SIM7000G → HiveMQ Cloud MQTT → backend ingestion service → Firebase → dashboard. The backend service subscribes to MQTT messages, validates them and writes records to Firebase. Document the implemented Firebase database service and ingestion configuration in the technical guide.

Area	Agreed scope
Testing	DHT11 temperature and humidity readings, TR-525M tip counting, ESP32 firmware and bench checks.
Final product	SHT31 temperature and humidity sensing, rain-gauge measurements and SIM7000G cellular communication.
Software	Firebase storage and access control; live and historical views, CSV export, station health and responsive dashboard access.
Outdoor preparation	Battery, charging, solar integration, voltage regulation, permanent assembly and enclosure preparation.
Open acceptance	Final installation, baseline field calibration and the 48-hour field test. Verify SHT31 implementation evidence before final acceptance.
Future enhancements	Pressure, wind and UV sensing, GPS reporting and over-the-air updates.

Success measures

Success means reliable tip counting, sensor readings with clear units and sensor identity, validated delivery into Firebase, usable dashboard views and evidence against the agreed acceptance criteria. Final deployment acceptance requires field results. Stakeholder feedback and completion dates must be recorded when available.

2. Team members and responsibilities

Member	Role	Responsibility
Redah Gamieldien	Project Leader	Requirements, hardware integration, coordination and acceptance decisions.
Angelo Adams	Frontend Developer and Designer	Dashboard layout, user experience, responsive views and frontend evidence.
Qaasim Isaacs	Backend Developer and Tester	MQTT ingestion, Firebase, authentication and access checks, integration and test evidence.
Ethan Williams	Analyst Researcher and Writer	Research, requirements, technical documentation and evidence organisation.
Rameez Karriem	Agile Lead and Presenter	Task tracking, meetings, reviews, retrospectives and presentation coordination.

3. Ways of working

We use a Kanban taskboard in Trello with weekly sprint planning. GitHub holds code, documents and evidence. Sprint tasks link to the product backlog and user stories. Task states are Product Backlog, In Progress and Completed.

Activity	Frequency	Purpose
Progress check-in	Three times per week	Discuss progress, next steps and blockers.
Planning	Weekly	Select tasks, assign leads and agree effort estimates.
Review or demo	End of each milestone	Inspect results and record feedback.
Retrospective	After each milestone	Agree an improvement and its owner.

Individual logbook entries cover Tuesday to Friday; these entries do not imply a formal stand-up on every day. Mondays fall outside this reporting schedule. Any required Monday attendance or work records must be handled according to the lecturer's instructions.

4. Working agreements and communication

We respect deadlines, communicate delays early, participate in meetings, complete assigned work and support one another. In-person discussion and WhatsApp are the main communication channels; Microsoft Teams supports remote meetings. The agreed response expectation is 20–30 minutes depending on urgency and availability.

We discuss decisions as a team and use a majority vote where needed. The Project Leader, acting as Product Owner for this project, makes the final decision if agreement cannot be reached. Changes to scope, sensor selection or data storage must be reflected in every affected document.

5. Definition of Done

- The task meets its acceptance criteria, and the actual output is linked to its task or story.
- A teammate has reviewed the work and the relevant tests have passed; faults and limitations are recorded.
- Code, wiring, configuration and documentation are updated consistently.
- Test records identify DHT11 or SHT31 so that testing and final-product results can be distinguished.
- Firebase records and dashboard displays use consistent station identifiers, timestamps and units.
- The Project Leader records acceptance. A document sign-off does not close installation or field validation.

6. Values and graduate attributes

Respect, transparency, accountability, trust and learning guide the team. We focus on user needs, deliver in manageable increments and use feedback to improve. The companion Graduate Attributes and Portfolio Evidence document applies Ubuntu, technological capability, resilience and ethical capability to this project and identifies the evidence each member must contribute.

7. Risks and dependencies

Risk or dependency	Response	Lead
Sensor or calibration inconsistency	Separate DHT11 test records from SHT31 final verification; retain reference measurements.	Redah / Qaasim
Cellular, MQTT or Firebase failures	Record connection and ingestion checks, access rules and recovery tests.	Qaasim
Outdoor power and exposure	Check charging, consumption, enclosure and installation before field acceptance.	Redah
Site access and field testing	Confirm access and baseline calibration before relying on a 48-hour result.	Redah
Missing evidence or uneven contribution	Link outputs, confirm individual records and raise blockers promptly.	Rameez / Ethan
Budget and skills constraints	Review component choices, costs and learning needs; retain actual records.	Whole team

8. Review and improvement

Review the charter at retrospectives and after significant scope changes. Record actual feedback, the agreed improvement, its owner and the next review date. Proposed improvements must not be presented as meetings that already occurred.

9. Goals and milestones

Our goal is to demonstrate an ESP32 rainfall prototype with remote data delivery and a usable dashboard within the recorded 13-week schedule. Measurable checks are defined in the user stories and sprint backlog. The current prototype scope includes temperature and humidity, data history, export, authentication and station-health information.

Period	Milestone	Recorded position
Week 1	Scope, components and architecture	Reported complete
Weeks 2–4	Firmware, cloud and cellular integration	Reported complete; final SHT31 evidence requires verification
Weeks 5–7	Dashboard, power and station diagnostics	Reported complete
Weeks 8–9	Assembly and deployment preparation	Assembly reported complete; installation and baseline calibration open
Weeks 9–10	Field validation	In Progress
Weeks 11–13	Documentation and presentation preparation	Prototype package reported complete; field acceptance remains open

The existing status snapshot runs to 23 October 2026: 43 original task cards are reported Completed and six deployment or field-validation cards remain In Progress. Completion labels require supporting evidence; they do not establish final field acceptance.

This records the supplied project schedule. The subject guide lists March to June for the eWIL assessment activities and July to November for WIL. The applicable assessment deadline and acceptance of this reporting period must be confirmed with the lecturer; the dates here are not a revised university deadline.

10. Team commitment

The original charter records the agreement initials below. They are retained as recorded agreements and are not new signatures or approval of this revision. The team should review the revised configuration and responsibilities together.

Member	Recorded agreement
Angelo Adams	A. Adams
Redah Gamieldien	R. Gamieldien
Qaasim Isaacs	Q. Isaacs
Ethan Williams	E. Williams
Rameez Karriem	R. Karriem